lab 10: Graphics in C

Objective

- To be familiar with file graphics in C programming

Program 1

Algorithm

Step1: start

Step2: Initialize Graphics node

Step3: Declare variable for starting point (x_1, y_1) and ending point (x_2, y_2)

Step4: Input values to (x_1, y_1) and (x_2, y_2)

Step 5: call the function to draw line between the two points

Step 6: Display the line on screen

Step 7: wait for user input

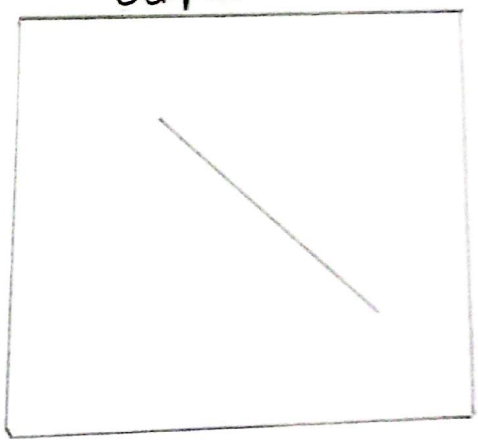
Step 8: Close graphic mode

step 9: stop

Cprogram

```
#include<stdio.h>
# include <graphics.h>
int main()
{
int gd=DETECT gm;
initgraph(&gd,&gm,"C:\\TURBO C3\\BGI");
line(100,100,300,300);
getch();
closegraph();
return 0;
}
```

Output



RESULT

The straight diagonal line was displayed carefully.

Program 3

Algorithm

Step1: start

Step 2: initialize graphics mode

Step 3: Declare variables for center (x_1, y_1) and radius (r)

Step 4: Input or assign values

Step 5: Call function to draw circle using center and radius

Step 6: Display the circle.

Step 7: wait for user input.

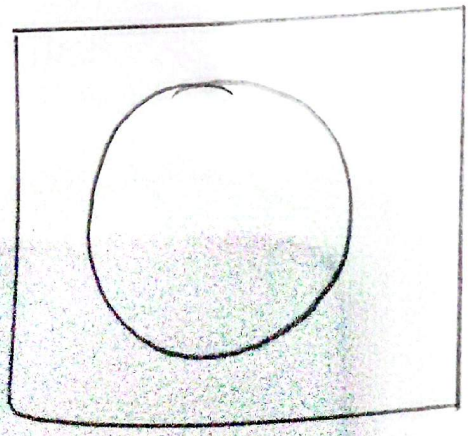
Step 8: close graphics mode

Step9: stop

C-program

```
# include <stdio.h>
# include <graphics.h>
int main ()
{
int gd = DETECT gm;
initgraph (&gd, &gm "C:\\TURBO C3\\BGI");
circle (200, 200, 100);
getch ();
closegraph ();
return 0;
}
```

Output.



RESULT

The proper circle is displayed carefully.

program 4

Algorithm

~~#inclu~~ step1: start

Step2: Initialize graphics mode

Step 3: Declare variable for center (x,y). horizontal radius (rx) and vertical radius (ry)

Step 4: Input or assign values

Step 5: call function to draw ellipse using parameters

Step 6: Display the ellipse

Step 7: wait for user input

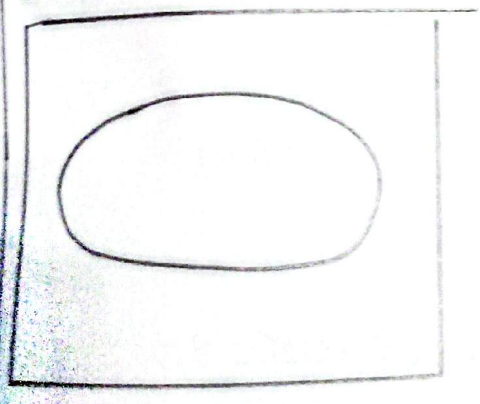
Step 8: close graphics mode

step 9: stop

C program

```
#include<stdio.h>
# include <graphics.h>
int main()
{
int gd=DETECT,gm;
initgraph(&gd,&gm,"C:\\TURBOC3\\BGI");
ellipse(250,200,0,360,150,80);
getch();
closegraph();
return 0;
}
```

Output



RESULT

The ellipse was printed successfully.

program 5

Algorithm

Step1: start

Step2: Initialize graphics mode

Step 3: Set text colour using colour function

Step 4: Set text style and font

Step 5: specify position (x,y) for text

Step 6: Display text on screen

Step 7: Wait for user input

Step 8: close graphics mode

Step 9: Stop

C-program

```
#include <stdio.h>
#include <conio.h>
#include <graphics.h>
int main()
{
int gd=DETECT gm;
initgraph(&gd, &gm, "C:\\TURBOC3\\BGI");
setcolor(BLACK);
settextstyle(BOLD_FONT, HORIZ.DIR, 3);
outtextxy(100, 200, "HELLO GRAPHICS");
getch();
closegraph();
return 0;
}
```

Output

HELLO GRAPHICS!

RESULT

Hello GRAPHICS:

CONCLUSION

In this lab session, we learned about different graphics in C.